

Do **not** write solutions on this page.

11. [Maximum mark: 17]

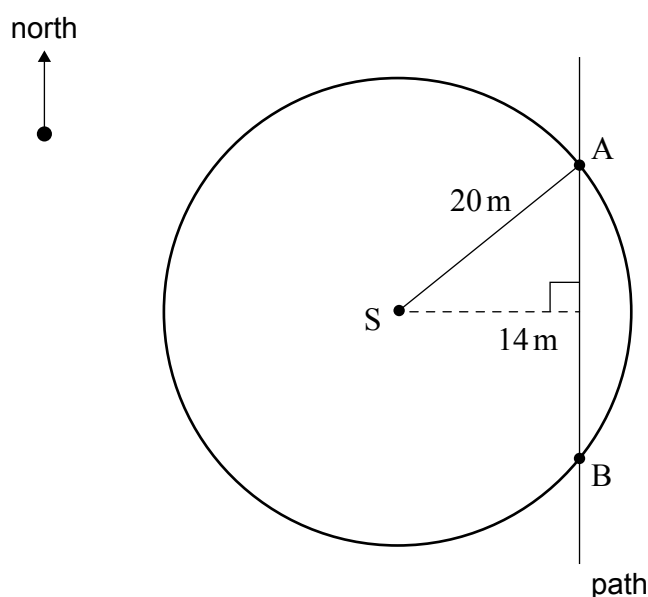
A rotating sprinkler is at a fixed point  $S$ .

It waters all points inside and on a circle of radius 20 metres.

Point  $S$  is 14 metres from the edge of a path which runs in a north-south direction.

The edge of the path intersects the circle at points  $A$  and  $B$ .

This information is shown in the following diagram.



(a) Show that  $AB = 28.57$ , correct to four significant figures. [3]

The sprinkler rotates at a constant rate of one revolution every 16 seconds.

(b) Show that the sprinkler rotates through an angle of  $\frac{\pi}{8}$  radians in one second. [1]

Let  $T$  seconds be the time that  $[AB]$  is watered in each revolution.

(c) Find the value of  $T$ . [4]

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**(Question 11 continued)**

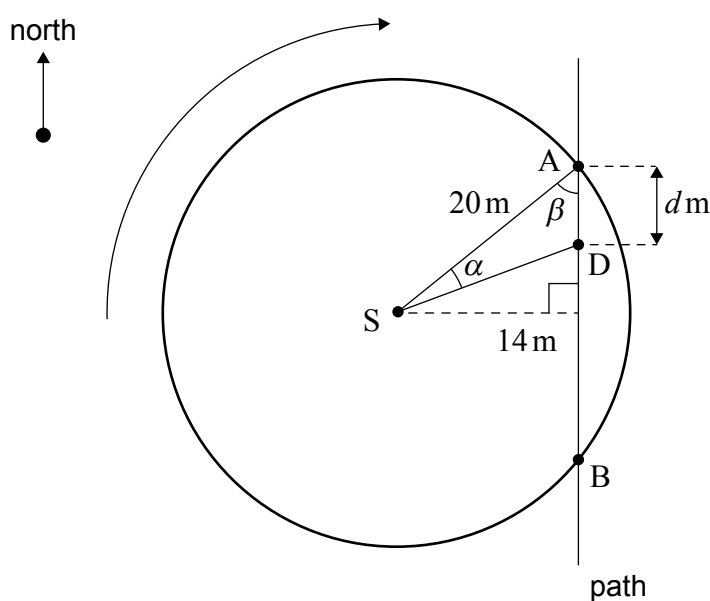
Consider one clockwise revolution of the sprinkler.

At  $t = 0$ , the water crosses the edge of the path at A.

At time  $t$  seconds, the water crosses the edge of the path at a movable point D which is a distance  $d$  metres south of point A.

Let  $\alpha = \angle ASD$  and  $\beta = \angle SAB$ , where  $\alpha, \beta$  are measured in radians.

This information is shown in the following diagram.



(d) Write down an expression for  $\alpha$  in terms of  $t$ .

[1]

It is known that  $\beta = 0.7754$  radians, correct to four significant figures.

(e) By using the sine rule in  $\triangle ASD$ , show that the distance,  $d$ , at time  $t$ , can be modelled by

$$d(t) = \frac{20 \sin\left(\frac{\pi t}{8}\right)}{\sin\left(2.37 - \frac{\pi t}{8}\right)}.$$

[3]

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**(Question 11 continued)**

A turtle walks south along the edge of the path.

At time  $t$  seconds, the turtle's distance,  $g$  metres south of A, can be modelled by

$$g(t) = 0.05t^2 + 1.1t + 18, \text{ where } t \geq 0.$$

- (f) At  $t = 0$ , state how far south the turtle is from A. [1]

Let  $w$  represent the distance between the turtle and point D at time  $t$  seconds.

- (g) (i) Use the expressions for  $g(t)$  and  $d(t)$  to write down an expression for  $w$  in terms of  $t$ .  
(ii) Hence find when and where on the path the water first reaches the turtle. [4]

